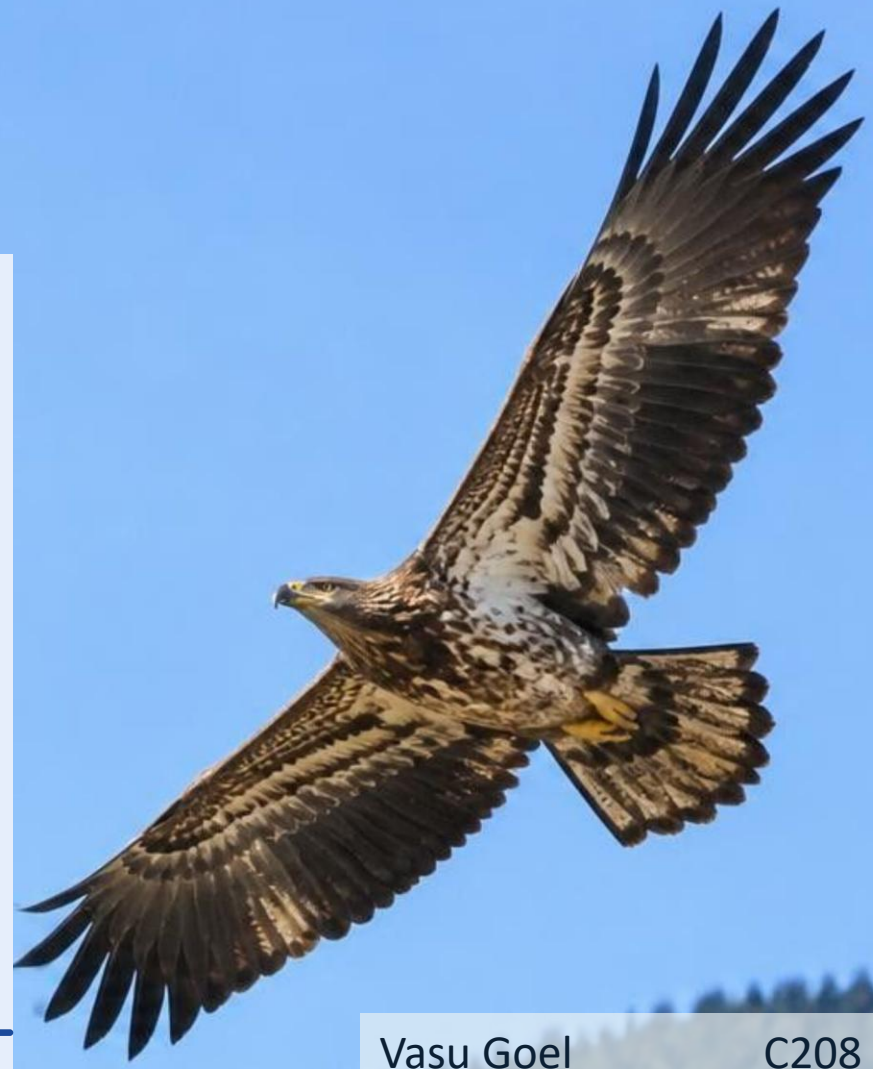
BIOMIMICRY · ELEMENTS OF BIOLOGY · B.TECH

From Nature to Engineering: Bird-Inspired Morphing Wings

How adaptive geometry connects avian flight, compliant structures and feedback control.

Biological inspiration → aerodynamic design → controlled shape change



Vasu Goel	C208
Naren Gounder	C209
Rachit Hegonde	C220
Parv Jain	C225

Conventional Aircraft

A Conventional Aircraft is a fixed-wing design in which different parts handle different jobs: the wings produce lift, the engines produce thrust, the tail provides stability, and movable control surfaces steer the aircraft.

Main characteristics

Separate functions: lift comes from the wings and thrust from the engines, so the two are independent of each other.

Fixed wings: the aircraft must move forward at speed to generate lift, so it needs a runway for takeoff and landing.

Tail at the rear: the horizontal and vertical stabilizers keep the aircraft steady in pitch and yaw.

Hinged control surfaces: ailerons (roll), elevators (pitch), rudder (yaw), and flaps and slats (extra lift at low speed) are all separate parts attached by hinges.

Many hinges and gaps: because these surfaces are separate from the fixed structure, there are gaps and joints along the wings, tail, and doors.

Conventional Aircraft





Problem Statements

- Aircraft Stalling
- Noise Levels
- Parasitic Drag
- Maintenance
- Fuel Burn
- Radar Signature

Aircraft Stalling

- Stalling is a sudden loss of lift. It happens when the wing is tilted too steeply into the airflow, and the air can no longer flow smoothly over the top of the wing.
- A stall is not the engine stopping. An aircraft can stall with the engines running at full power.
- **Angle of attack (AoA):** angle between wing's chord line (the straight line from the front to the back of the wing) and direction of the oncoming air.

Lift is given by:

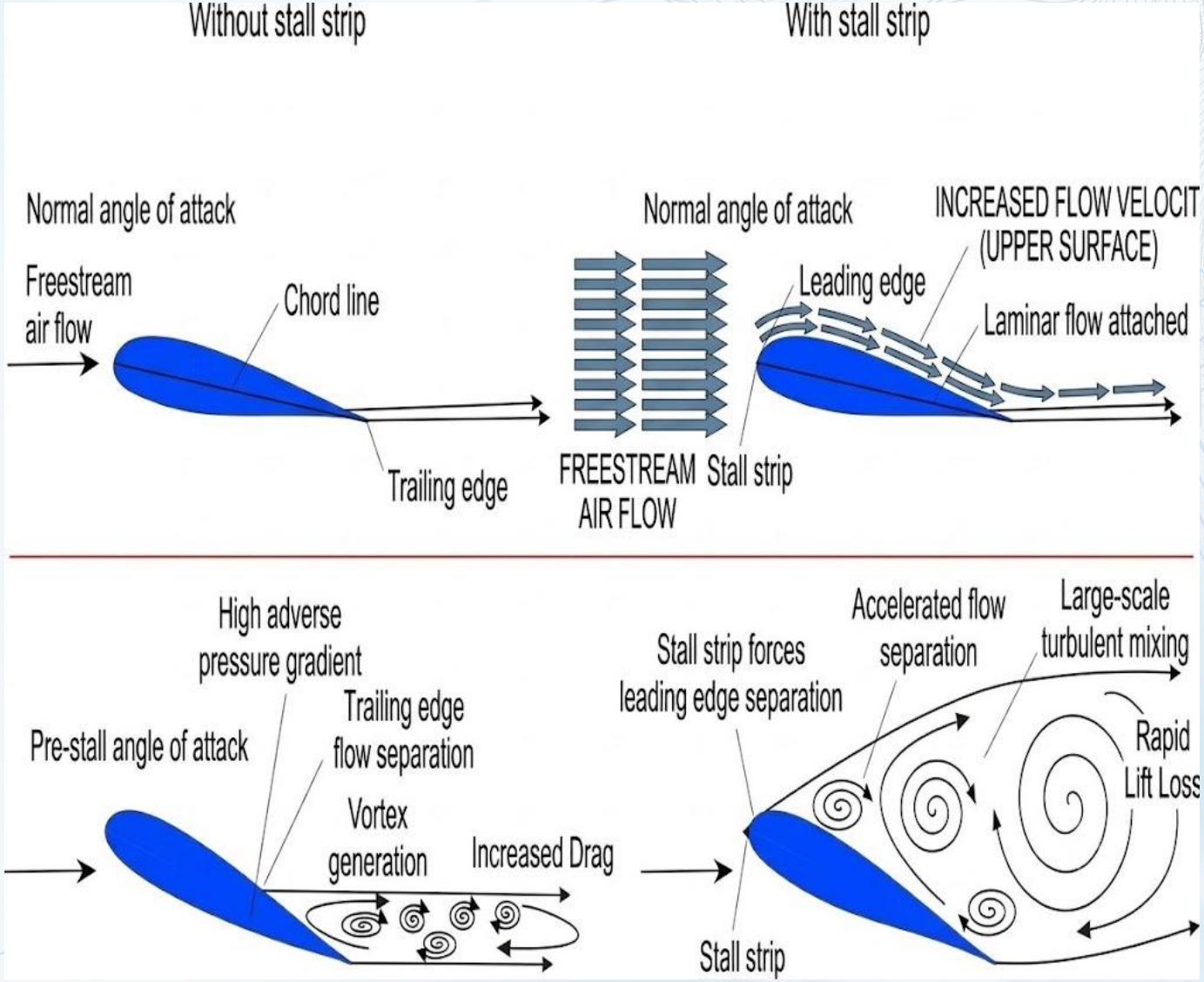
$$L = \frac{1}{2} \rho V^2 S C_L$$

where

- ρ is air density,
- V is speed,
- S is wing area and
- C_L is the lift coefficient (which depends on the angle of attack).

How stall happens

- At a small angle, air flows smoothly over the wing, create lifts. Lift increases → angle increases.
- When the angle becomes too high (critical angle), airflow stops flowing smoothly (separate from upper surface of wing)
- Wing produces less lift and air resistance increase flow
- Aircraft may start losing its height due to wing can not produce enough lifts to support its weight



Noise Levels

Where the noise comes from

- **Gaps and edges:** Air passes thru the gaps between flaps and wings, this create noise
- **Flap side edges:** Air moves around the slides of the flaps and it creates noise
- **Trailing edge noise:** Air flows over the wing becomes disturbed near its back edge. This creates noise, like a continuous hissing sound
- **Landing gear and cavities:** Not flaps, the landing gears and open spaces in the aircraft also create noise during landing

Parasitic Drag

Drag means air resistance and parasitic means unwanted situation which consumes more energy

Parasitic drag is the air resistance an aircraft faces while moving through the air. It makes the aircraft use more energy.

- Skin Drag **:** Air rubs against the aircraft's surface and slows it down.
- Form Drag (Pressure Drag) **:** The shape of the aircraft creates air resistance.
- Interference Drag: When air flows around two parts that meet, like the wing and body, extra drag is created.

The formula

$F_D = \frac{1}{2} \rho v^2 C_D A$

The key point is that parasitic drag grows with the **square of speed**. Doubling the speed makes parasitic drag four times larger.

Maintenance of conventional aircraft

- **Many Moving Parts**: Flaps and slats have many parts, such as hinges, rollers, tracks, and motors. These parts need regular checking and replacement.
- **Wear and Tear**: Metal parts rub against each other whenever flaps open and close. Over time, this causes the parts to become loose or damaged.
- **Jamming**: Dust, ice, or damaged parts can stop flaps or slats from moving properly. If the flaps on both wings move differently, the aircraft may tilt to one side.
- **Corrosion**: Rain, moisture, and other fluids can cause metal parts to rust and become weak over time.
- **Lubrication**: Moving parts need regular greasing to reduce friction and prevent damage.
- **Fatigue**: Repeated stress during flights can cause small cracks in parts such as tracks, brackets, and joints.
- **Regular Inspections**: Engineers regularly check these parts, look for cracks, test flap movement, and replace damaged or worn-out components.

Fuel burn and Radar signature

Fuel Burn

- Extra Weight: Hinges, tracks, rollers, and motors add weight to the aircraft. A heavier aircraft needs more lift, which increases drag and fuel consumption.
- Fixed Wing Shape: Conventional wings are designed to work best under certain conditions. In other conditions, they may create more drag and use more fuel.
- Flap Deployment: When flaps are extended during takeoff and climbing, they increase drag. This can reduce climbing performance and increase fuel consumption.

Radar Signature: How Hinges and Gaps Reflect Radar

- Edges and Gaps: Sharp edges and gaps on the aircraft scatter radar waves in different directions. Some waves travel back toward the radar, making the aircraft easier to detect.
- Hinges and Corners: Hollow spaces around hinges and gaps can reflect radar waves back toward their source. This can make the aircraft more visible to radar, similar to how corner reflectors help boats appear on radar screens.

BIOLOGICAL PHENOMENON

- OWL

FEATHERS FOR SILENT FLIGHT

- Owls have specialized feathers that allow them to fly almost silently while hunting. Their primary feathers feature comb-like serrations, soft surfaces, and fringed edges that help reduce aerodynamic noise.

- Inspired by these adaptations, engineers study serrated edges and owl-inspired structures to reduce noise around aircraft wings, engines, and fan blades. This demonstrates how nature can inspire quieter and more aerodynamically efficient designs.



AIRCRAFT WINGS VS OWL WINGS

Comparing wing-edge design, airflow and aerodynamic noise





Conventional Aircraft Wing

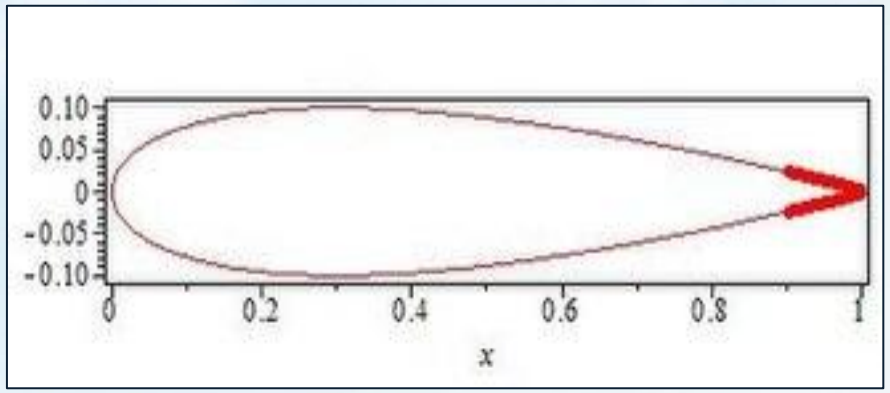
- Rigid metal or composite surface with a smooth, straight and sharp trailing edge.
- Air from the top and bottom surfaces meets abruptly at the edge, forming large and organised vortices.
- These vortices create strong pressure fluctuations, which are heard as aerodynamic noise, mostly during approach and landing.

Owl Wing

- Flexible feathers with a serrated leading edge, a soft velvety upper surface and a fringed trailing edge.
- The comb-like front splits incoming air into many small streams, and the fringe lets air from above and below mix gradually.
- Large vortices break into small ones, so pressure changes are smoother and the flight is almost silent.

Trailing Edge & Feather Structure

WHY THE SHAPE OF THE EDGE DECIDES HOW MUCH NOISE IS MADE



Cross section of an aerodynamic surface with the trailing edge emphasised



Comb-like leading edge of Barn Owl flight feather.

Aircraft trailing edge (close-up)

- Straight, hard edge with no flexibility to absorb pressure changes.
- Turbulent air reaching the edge scatters sound, known as trailing-edge noise.
- Serrated or porous edges are being tested as quieter alternatives.

Owl feather structure

- Overlapping feathers with fine, flexible barbs at the tips.
- Porous, soft structure absorbs pressure fluctuations.
- Fringes break up turbulence before it leaves the wing.

➢ SHARK

Wingtip Design for Reduced Drag

- Sharks have streamlined bodies and fins that help them move efficiently through water.
- Inspired by the shape of a shark's dorsal fin, Airbus developed **Sharklets**, upward-curving wingtip devices used on A320-family aircraft.
- These extensions help manage airflow around the wingtips and reduce induced drag caused by wingtip vortices.
- Reduced drag improves aerodynamic efficiency and can lower fuel consumption.
- Sharklets demonstrate how nature-inspired shapes can influence aircraft design and improve flight performance.
- **This highlights the role of aerodynamic design in making air travel more efficient.**





SPECIAL FOCUS

MORPHING WINGS

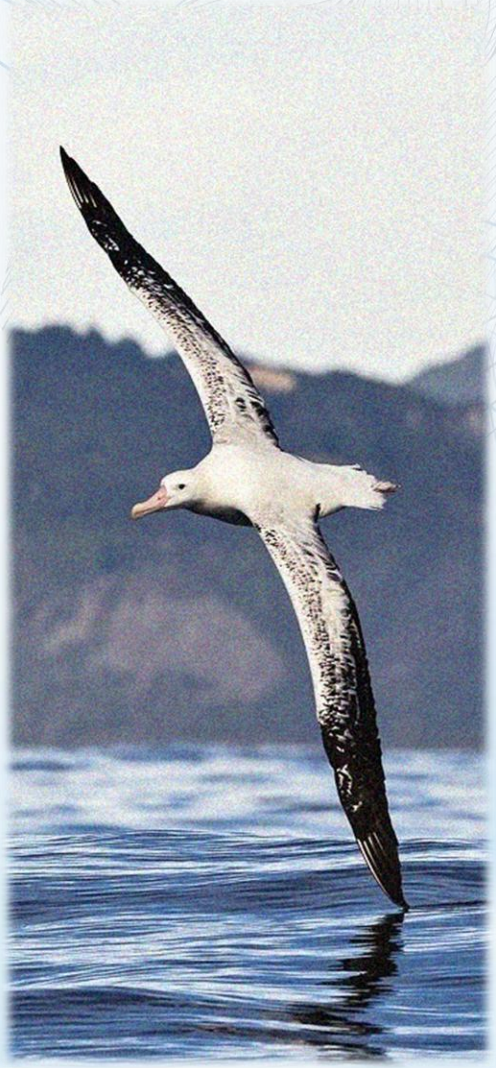
A bird's wing is basically an arm and a hand covered in feathers — and it changes shape all the time.

- **Arm-wing** (shoulder to wrist) holds the secondary feathers; **hand-wing** (wrist to fingers) holds the primaries.
- In pigeons, just two joints — the wrist and a finger — move all 40 flight feathers.
- Tiny hooks on the feathers lock them together when the wing opens and let them slide apart when it folds.



➢ ALBATROSS

Wings for Endurance



The wandering albatross is a seabird known for travelling enormous distances over the ocean with very little wing flapping. Its specialized wings and **dynamic soaring** technique help it achieve energy-efficient flight.

- **Long, Narrow Wings:** Its exceptionally long, slender wings have a high aspect ratio, helping reduce induced drag during gliding.
- **Efficient Gliding:** Its wings generate lift, allowing it to remain airborne for long periods without continuous flapping.
- **Dynamic Soaring:** The albatross changes its direction and height to extract energy from differences in wind speed at different altitudes.
- **Extended Wings and Flight Feathers:** Its large wingspan provides an effective lifting surface, making it well adapted for sustained soaring over open water.

➢ SWIFT

Wings for Agile Flight

- Swifts are specialized aerial birds with long, narrow, sickle-shaped wings that enable fast and agile flight.
- Their flexible wing joints allow them to adjust wing sweep and posture according to flight speed and manoeuvring needs.
- When gliding or turning, they extend their wings to generate lift.
- At higher speeds, sweeping their wings backward changes their aerodynamic performance.
- These adaptations help balance lift, drag, speed, and manoeuvrability.
- The swift inspires research into **variable-sweep and morphing wings** that adapt to different aircraft flight conditions.





PEREGRINE FALCON

Wings for High-Speed Flight

- The peregrine falcon is renowned for its high-speed hunting dive, known as a **stoop**

- During the dive, it draws its wings closer to its body, creating a streamlined shape that reduces frontal area and helps achieve high speeds.

- During turns and manoeuvres, it adjusts its wing and tail positions to control lift, drag, and direction.
- These adaptations help balance speed, stability, and manoeuvrability.

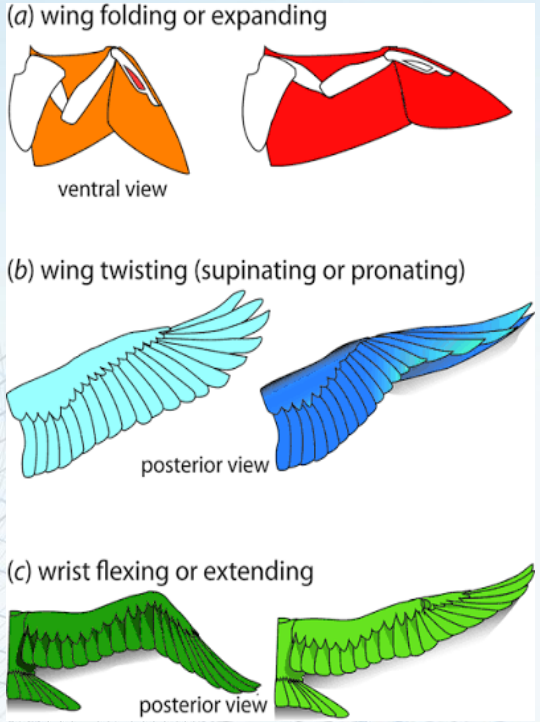
- The falcon demonstrates adaptive aerodynamics and inspires research into **morphing aircraft wings** that adjust to different flight conditions.

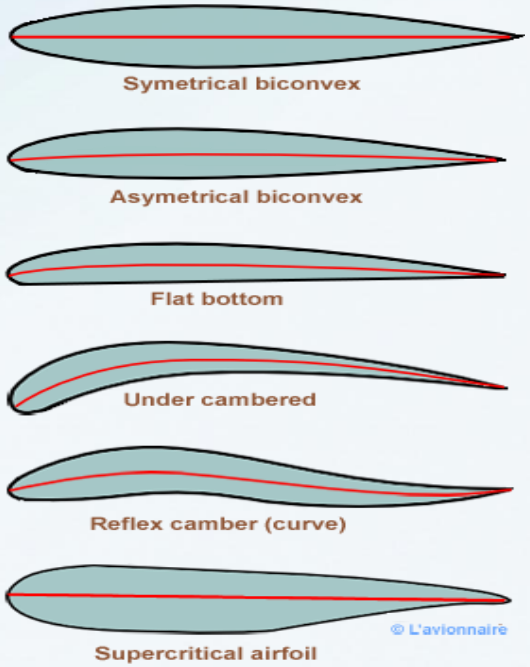




BIOLOGICAL INSPIRATION

Birds reshape their wings in flight, and aircraft are learning to do the same the same





In flight, birds reshape wings and tails to control pitch, roll pitch, roll and yaw (rotation about the aircraft's lateral, lateral, longitudinal and vertical axes)—morphing. Joint Joint motion and feather adjustment change geometry with geometry with speed and manoeuvre demands.

A fixed wing is a compromise across flight conditions, even at its preferred operating point. Avian-inspired flight seeks to transfer this adaptability to unmanned aerial vehicles (UAVs), but comparably sized UAVs have so far not matched birds' manoeuvrability. Appearance alone cannot reproduce integrated mechanics and control; morphing bridges biology and engineering.

Research context: avian morphing study available through NSF Public Access Repository, Repository, par.nsf.gov.


nature

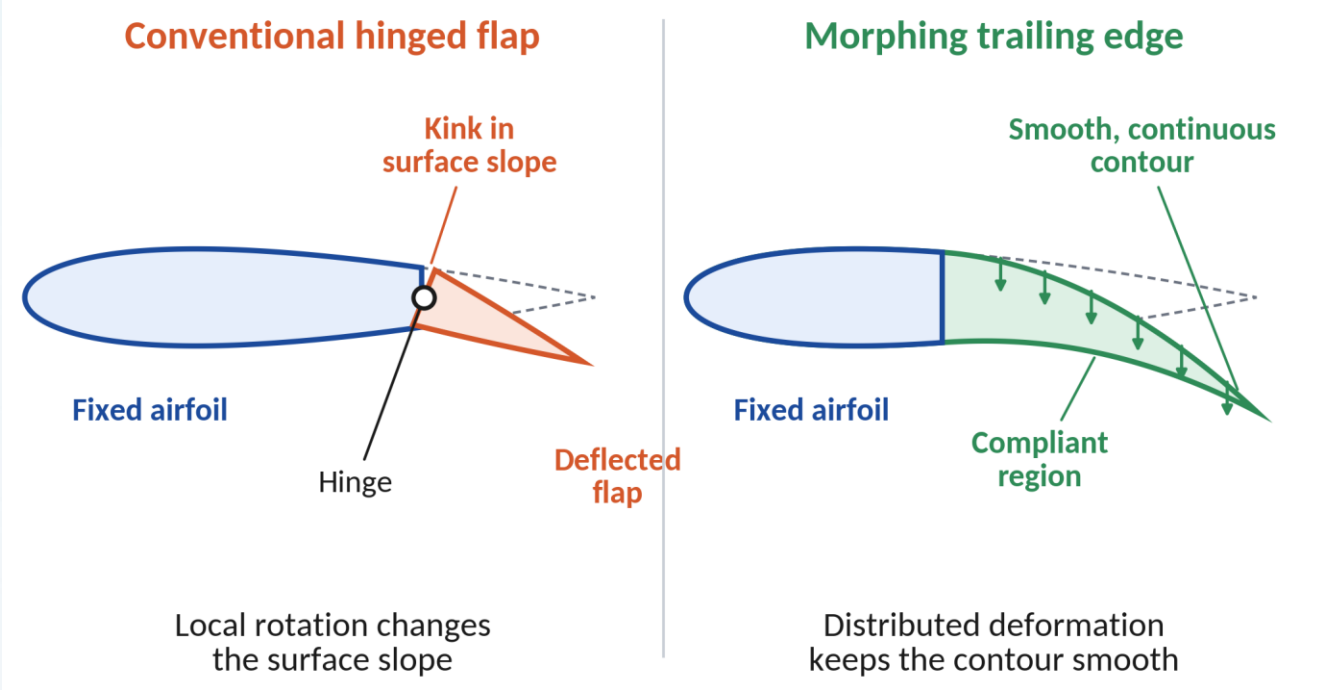
DEFINITION

What is a morphing wing?

- Dynamic Adaptation: Morphing wings adjust their shape in real time to match changing flight conditions.

- Continuous Deformation vs. Hinged Design: Unlike conventional rigid flaps and ailerons that pivot on hinges and create kinks in the wing profile, morphing surfaces distribute shape changes smoothly and continuously.

- Performance Benefits: Designed to deliver greater aerodynamic efficiency, extended flight range, and improved manoeuvrability within a single airframe.



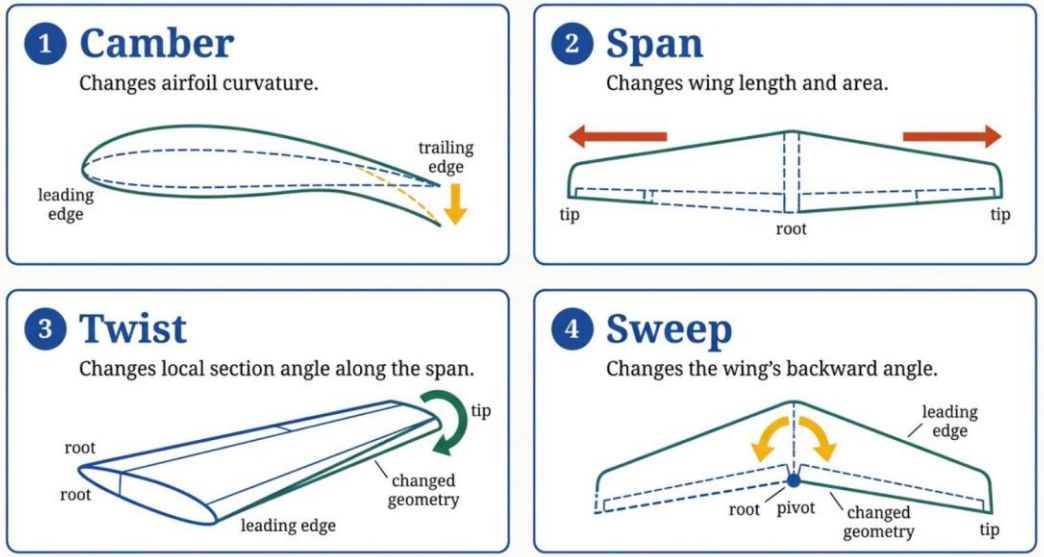
Definition and adoption context: morphing-wing review, University of Southampton repository, repository, eprints.soton.ac.uk.

GEOMETRY

Four ways a wing can morph

Morphing changes a wing's cross-section, planform, or the orientation of sections along its length. Planform is the outline seen from above. These are distinct geometric changes, not interchangeable mechanisms, and a practical design may combine them.

Camber modifies section aerodynamics; span changes lifting-surface extent; twist redistributes aerodynamic loading; sweep changes how the wing meets the flow. Each affects more than one performance variable. Birds also adjust dihedral—the upward inclination of wings—and tail geometry, extending the biological design space beyond these four modes.



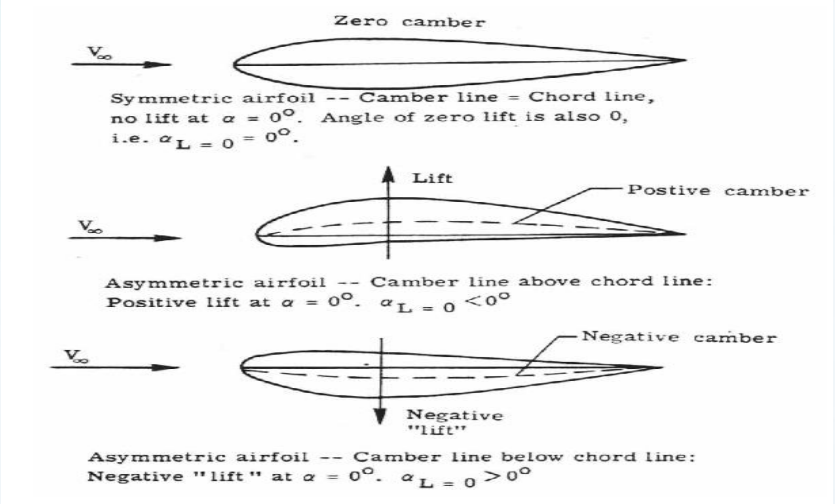
Biological context: NSF Public Access Repository, par.nsf.gov. Geometry definitions follow standard aerodynamics.

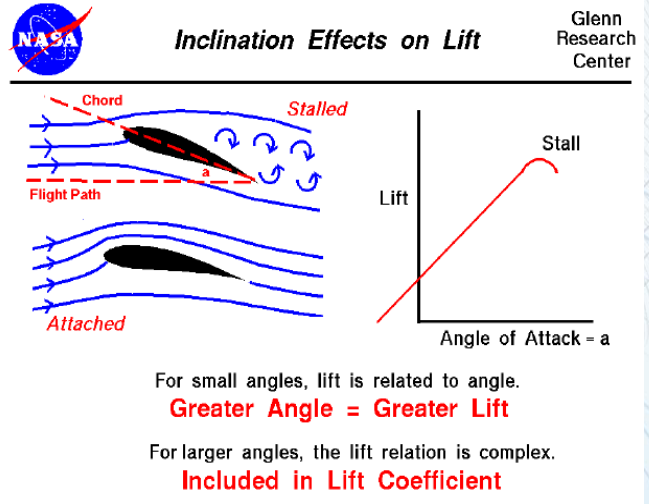
AIRFOIL MORPHING

Variable camber: changing the airfoil

Camber is the curvature of an airfoil's mean line; more camber raises lift at a given angle of attack, the angle between the chord and incoming airflow.

Continuous trailing-edge designs such as FishBAC (Fish Bone Active Camber) replace hinged flaps.





Up to 25%

Lift-coefficient increase

Up to 37%

Drag reduction

Peak values from FishBAC wind-tunnel tests (rear 25% of the chord the chord flexing up to 32°). A separate test found over 50% higher higher lift-to-drag than a plain flap at moderate-to-high lift.

Sources: Rakhshani et al., FishBAC wind-tunnel study (peak lift/drag); Rivero et al., City, Univ. of London and Bristol (lift-to-drag vs plain flap); morphing-wing review, iieta.org. review, iieta.org.

WING-LEVEL MORPHING

Beyond camber: span, twist and sweep

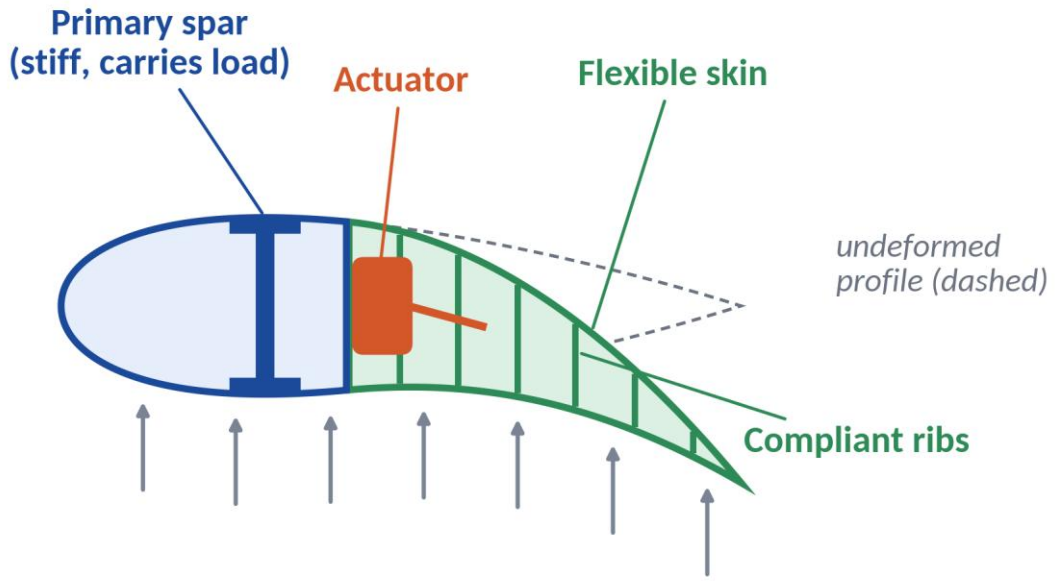
- **Span Extension and Retraction:** Extending the wing span can reduce induced drag (the drag associated with producing lift), though it increases structural demands; conversely, retraction reduces wing area to suit faster flight.
- **Wing Twist:** Modifying the twist changes the local angle of attack; applying unequal twist between the left and right wings can provide roll control without relying on conventional ailerons.
- **Wing Sweep:** Sweeping the wing moves it backward to optimize for high-speed flight, which serves as an analogy to bird wing folding, though the aerodynamics are not identical.
- **Stability and Integration Impact:** Every morphing degree of freedom directly affects the aircraft's stability and control; the ultimate benefits depend on the specific flight regime and the design of the complete airframe.

Mode	Benefit	Trade-off
Span	Efficient cruise	Greater bending demand
Twist	Load shaping and roll	Coupled deformation
Sweep	Fast-flight suitability	Stability changes

Avian stability and control context: par.nsf.gov. Benefits and trade-offs follow standard aerodynamics.

STRUCTURES

The structural challenge



Flexible to move.
Stiff to carry load.

The skin must change shape without losing its aerodynamic contour under pressure. Compliance Compliance means controlled elastic deformation deformation rather than unrestricted softness.

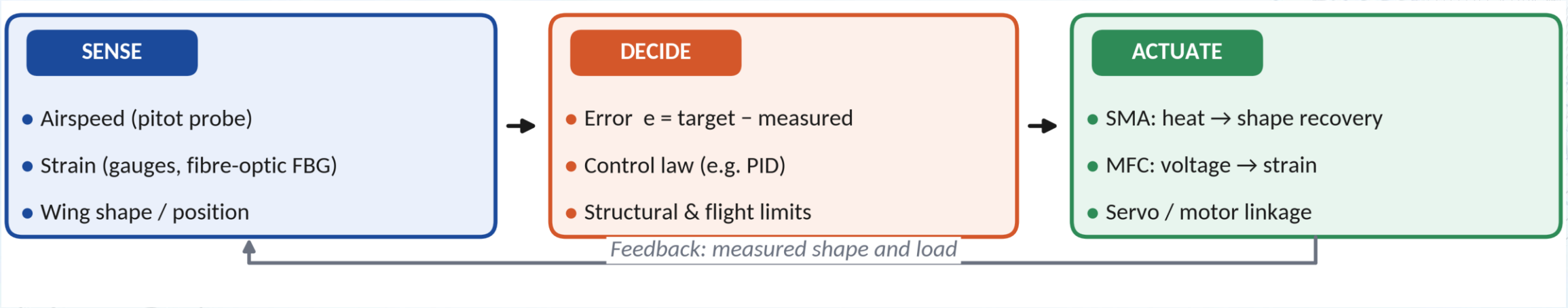
A spar is the main spanwise load-carrying member. Compliant ribs guide shape change, while flexible skins skins preserve a continuous surface.

These requirements compete. Actuators must respond quickly enough and overcome aerodynamic loads. Hysteresis means the response depends on loading depends on loading history, complicating accurate positioning. Fatigue limits repeated deformation. Aeroelastic coupling links airflow, structural motion and structural motion and aerodynamic force, potentially changing stability. Scaling a successful laboratory mechanism to a larger aircraft therefore requires more therefore requires more than enlarging its dimensions.

Open challenges: experimental morphing-wing review, iieta.org.

MECHATRONICS

Sensors, actuators and control



Sense

Sensors measure airspeed, strain and wing shape. Strain is deformation relative to original length; measurements show whether the wing has reached its target geometry under current load.

Decide

The flight controller compares the actual shape of the wing and target shape required. This control error guides actuator commands within structural and flight limits.

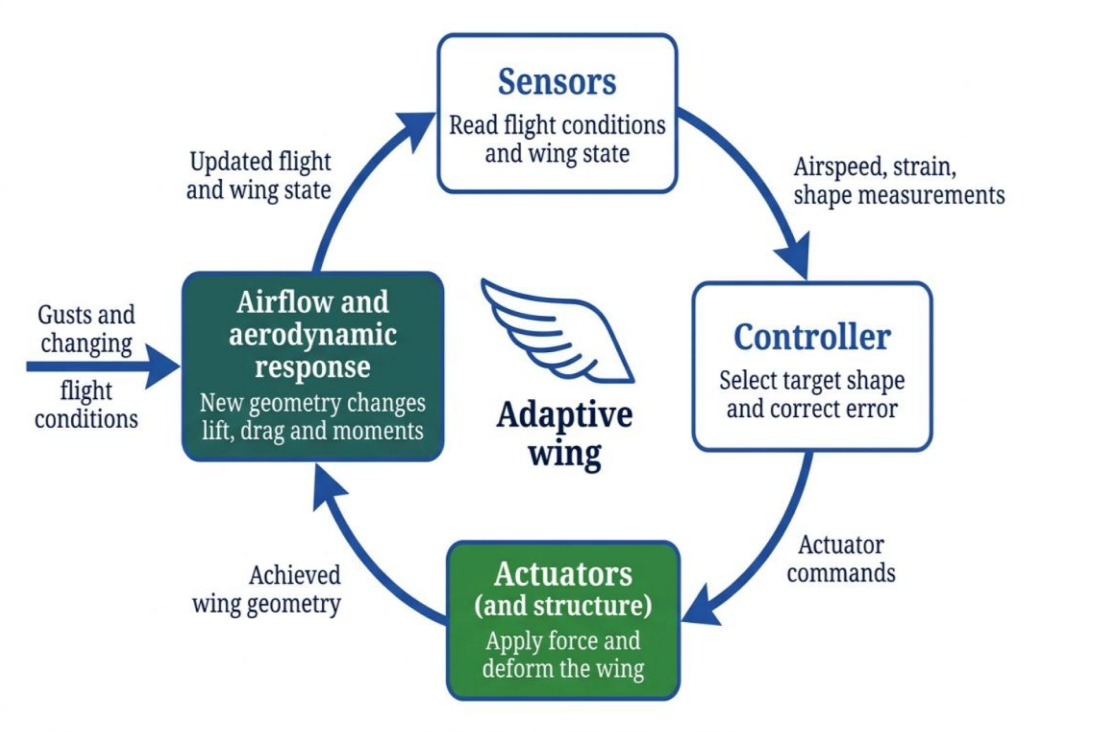
Actuate

Shape memory alloys (SMAs) recover a trained shape when heated. Macro-Fiber Fiber Composites (MFCs) use piezoelectric piezoelectric fibres for electrically driven driven strain; their force, displacement and and response differ.

Actuator examples: experimental morphing-wing review, iieta.org. The architecture illustrates a feedback-controlled system.

SYSTEM INTEGRATION

The complete working principle



Signal flow → command → structural deformation → aerodynamic response → → feedback.

Closed-Loop Architecture: Operates on continuous feedback—measuring dynamic wing outputs to dictate subsequent control commands under shifting aerodynamic loads.

Intelligent Shape Selection: The onboard controller calculates target geometry based on real-time flight conditions, aerodynamic models, and structural constraints.

Compliant Actuation: Actuators introduce direct force or strain into the compliant airframe, reshaping the surface to dynamically alter airflow, lift, drag, and moments.

Real-Time State Verification: Sensor arrays track actual wing deformation against the commanded profile, closing the loop to ensure structural safety and optimal tracking.

TECHNOLOGY TAKEAWAYS

Technology takeaways

- ## Adaptive geometry

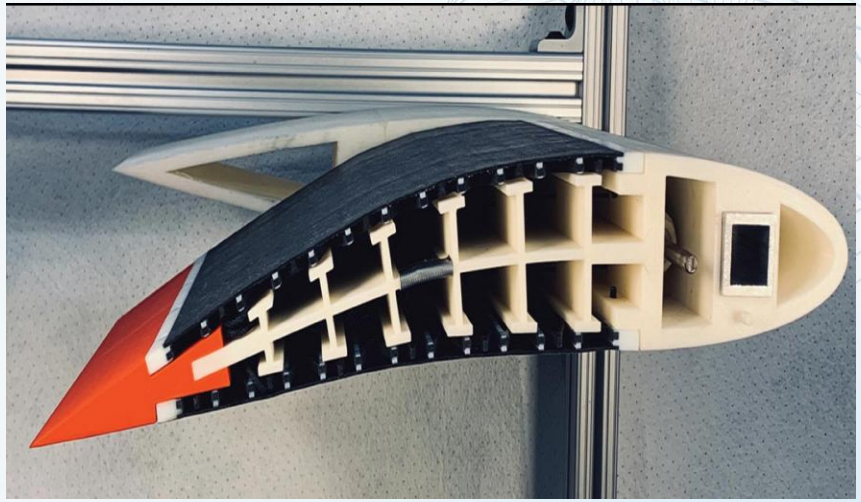
Birds show how continuous shape change adapts geometry to varied flight varied flight demands, unlike fixed geometry.

- ## Tested aerodynamic gains

Wind-tunnel tests of a FishBAC wing report up to 25% higher lift coefficient coefficient and up to 37% lower drag.

- ## Adoption depends on integration

Structures, actuators and certification still limit adoption; next step: smart materials and integrated control.



Sources: par.nsf.gov (avian morphing) · eprints.soton.ac.uk (morphing-wing review) · iieta.org iieta.org (actuator review) · Rakhshani et al. (FishBAC wind-tunnel data).



Where Morphing Wings Have Been Put to Work

Five distinct program families, spanning experimental testbeds, tactical fighters, and commercial fleets.

DOMAIN / TIER	PROGRAM & AIRFRAME	KEY MECHANISM & FLIGHT RESULTS	FLIGHT STATUS
Research Jet	**NASA ACTE**	Seamless flaps; 23 research flights with all test objectives verified.	**Flight-Tested**
Scaled Demos	**Airbus AlbatrossONE & X-Wing**	Freely-hinged wingtips deployed in flight to passively shed gust loads.	**Demonstrator**
Fighter Testbed	**X-53 Active Aeroelastic Wing**	Up to 5° wing twist utilized for transonic roll control without hinges.	**Flight-Tested**
Tactical UAV	**DARPA MFX-1 & MFX-2**	Span expanded +73% and aspect ratio +177% dynamically in flight.	**Flight-Tested**
Airliners	**Airbus A350 & Boeing 787**	Variable-camber flaps continuously trimmed in cruise to optimize lift.	**In Service**

NASA ACTE: a seamless flap flown on a Gulfstream III



Credit: AFRL / NASA. Source: AFRL release "Flight tests conclude for shape changing aircraft flap", wpafb.af.mil; NASA ACTE page, nasa.gov/centers/armstrong/research/ACTE

+30°
max flap deflection flown (from −2°)

Mach 0.85
top speed reached in ACTE phase II

Up to 30%
less airframe noise at takeoff and landing (NASA-reported)

4% to 8% reduction in fuel consumption

- **Program:** AFRL, NASA and FlexSys replaced each wing's Fowler flap with a 19 ft FlexFoil compliant flap.
- **No hinges:** distributed compliance bends the structure itself, so the surface stays seamless.
- **Result:** all flight-test objectives met in 23 ACTE-I flights, with 5,800+ sensors on the aircraft.
- **Limit:** lift gain flattens above ~15° as flow separates; the flap is ~4% heavier than the original.

Sources: NASA NTRS 20160007777, 20150019388, 20180005456; Kota & Martins, ICAS 2018 (FlexSys / U. Michigan); AFRL release.

Airbus: AlbatrossONE to the eXtra Performance Wing

2019–20 AlbatrossONE (scaled RPV)

- **Hinge:** an outward-flared wingtip hinge lets the tip fold freely and shed gust loads.
- **Flights:** free-folding tips were statically and dynamically stable, and recovered after gusts.
- **Watch-out:** past the flutter boundary, tips settle into a bounded oscillation set by flare angle.

2023–26 eXtra Performance Wing (Citation VII)

- **Scale:** 16 m span, about one-third of the >50 m target wing.
- **Status:** baseline flight 6 Nov 2023; new composite wing fitted in 2025.
- **Next:** gust-reactive tabs and release hinges; remote-piloted flights targeted late 2026.



Credit: Airbus. Source: airbus.com/en/newsroom/stories/2023-11-extra-performance-wing-demonstrator-takes-off

5–10% fuel-burn gain targeted by Airbus UpNext

Sources: Wilson et al. (Airbus), AlbatrossONE papers; AIAA J. Aircraft 10.2514/1.C037390; Airbus newsroom; Aviation Week; FlightGlobal; Aerospace Testing Intl.

Military and UAV programs: morphing in flight

Manned fighter: X-53 Active Aeroelastic Wing



Credit: NASA Armstrong. Source: nasa.gov/aeronautics/x-53/

- **Concept:** a thin-skinned F/A-18 wing, twisted up to 5° by its own surfaces, rolls the jet.
- **Result:** roll rates near a production F/A-18 at transonic and supersonic speeds.
- **Flown:** 2002–05 by NASA, AFRL and Boeing.

UAVs: DARPA Morphing Aircraft Structures

+73%

span change, MFX-2

+177%

aspect ratio change, MFX-2

- **First morph:** the 45 kg MFX-1 changed wing shape in flight on 1 Aug 2006, in about 5–10 s.
- **Agility:** MFX-2 turn radius differed by more than 2× between extreme shapes.
- **Setback:** Lockheed Martin's Z-wing UAV was dropped after two take-off crashes, which DARPA put down to flight-control software.

Sources: NASA X-53 / AAW pages and NTRS 20050203672; Flight International (2006–07) on MFX-1 and MFX-2; AFCEA SIGNAL; Lockheed Martin MAS wind-tunnel report, NTRS 20070018202.

Commercial aviation: variable camber flies today

IN SERVICE

Variable camber on the A350 XWB and 787

- A350 flaps and spoilers droop in cruise to tailor camber and relieve loads.
- A350 and 787 were the first airliners to trim flaps and ailerons in cruise for fuel burn.
- Hinged surfaces, not seamless skins: a stepping stone to full morphing.

ON NEW DESIGNS

Folding wingtips for longer spans

- Boeing 777X folds its tips to meet airport gate limits.
- Airbus' hinge goes further: tips are released in flight to shed gust loads.
- A longer span means more lift and less drag, per Airbus.

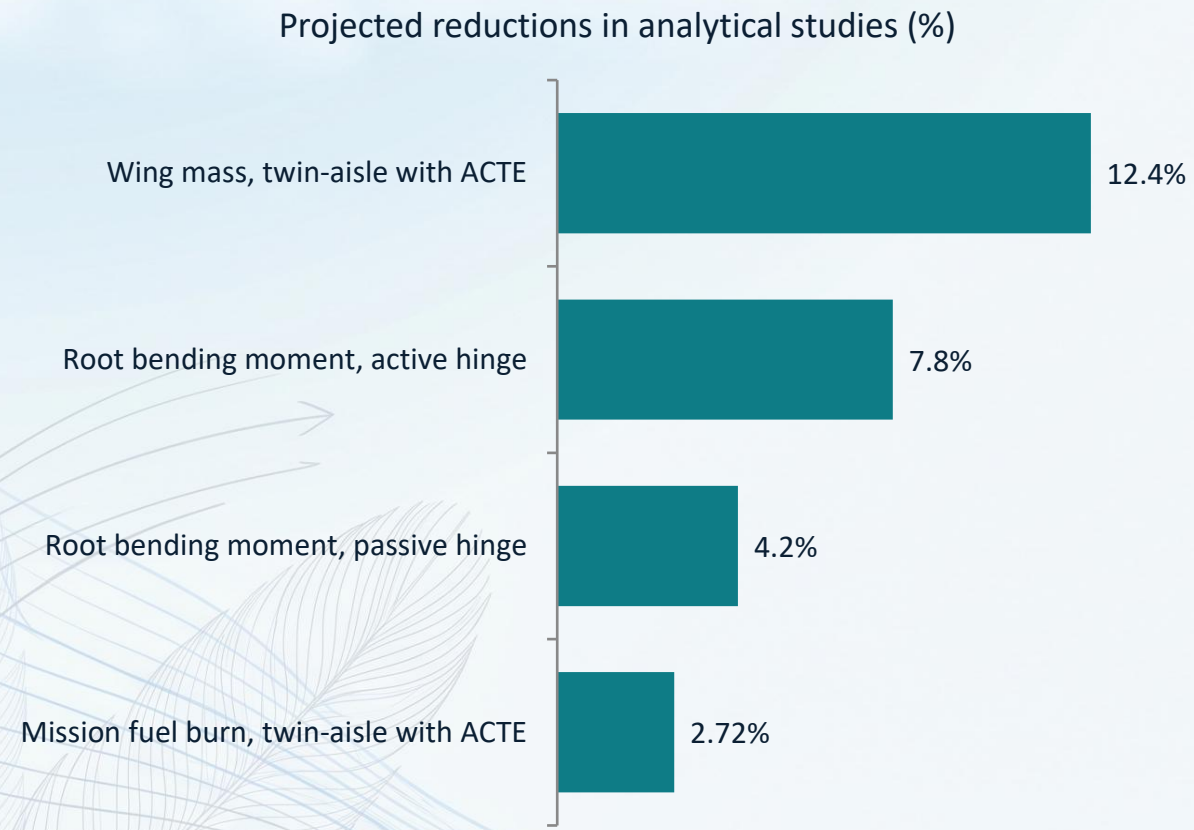
IN DEVELOPMENT

Airbus Wing of Tomorrow and ACTE-type surfaces

- Over 100 wing technologies studied in Wing of Tomorrow (Airbus, 2023).
- A321neo flight campaign for high-aspect-ratio wings reported July 2026.
- FlexSys proposes ACTE-type surfaces for trim tabs, winglets and ailerons.

Sources: Airbus A350 high-lift paper, ICAS 2014; Reckzeh (Airbus), cited in TU thesis and Aerosp. Sci. Technol.; FlightGlobal; Airbus; Aviación al Día (Jul 2026); Kota & Martins 2018.

What the evidence shows, and what it does not



Computed, not flight-measured. ACTE rows: CFD and structural optimization of a clean-sheet twin-aisle airliner. Hinge rows: aeroelastic study of a Cessna Citation X wing.

1. **Weight and power.** The ACTE flap weighed ~4% more than the flaps it replaced and needs 10–20% more drive power.
2. **Low-speed lift.** Seamless flaps lose slots and area; flow separated above ~15°, so flow control is needed.
3. **Prediction.** Pre-flight CFD over-predicted ACTE lift, so models still need flight data.
4. **Scale.** Flight evidence comes from business jets, subscale RPVs and one fighter, not an airliner.

Sources: Kota & Martins, ICAS 2018; AIAA J. Aircraft 10.2514/1.C037390; NASA NTRS 20160007777; Airbus AlbatrossONE papers.


ST
IAS
VIATIONWEEK

Pros & cons of morphing wings

Advantages	Limitations & challenges
• ≈7% less cruise drag (F-111 MAW flights)	• 15° ACTE flap loads levelled off above
• 10–20% lighter wing structure (X-53)	• 40° max sweep to land the X-5
• Variable camber gives a 1–3% gain in cruise, over 10% off-design (NASA study)	• Actuator weight and complexity “hindered acceptance” of the F-111 MAW (NASA)
• Same F-111 flights: over 20% less drag off-design, where fixed wings are weakest	• Shape change alters stability: one Bell X-5 was lost in a spin in 1953
• Gapless flaps reduce airframe noise at takeoff and landing (NASA ACTE)	• Skins must stretch yet hold shape under load; routine use is “still scarce”

The trade-off: it pays off only if the drag saved outweighs the extra mass, power and upkeep.

`Sources: Miller et al., NASA NTRS 20160000937 · Bolonkin & Gilyard, NASA/TM-1999-206586 · NASA X-53 · NASA X-5 · Li et al., Prog. Aerosp. Sci. 100 (2018) · NASA ACTE

Research timeline



Wright glider, 1902
1902

1900s
Flexible beginnings
Early pilots twisted the whole wing to roll and turn.



Douglas DC-3

1920s–40s
The rigid wing
Stiff metal wings with hinged flaps flaps become the standard.



Bell X-5

1950s–70s
Wings that sweep
Jets swing their wings back in flight for high speed.

Wings began flexible, turned rigid, and are learning to bend again.



NASA lattice wing

2020s →
Wings that adapt
Smart materials and sensors help help wings adapt on their own.



NASA test jet

2010s
Seamless surfaces
Gapless flaps are flight-tested; airliners adjust camber in cruise.



AFTI/F-111

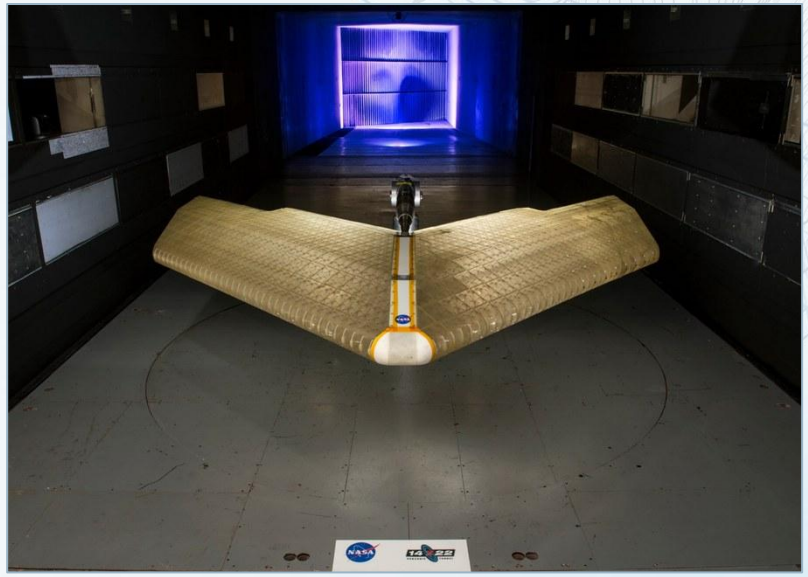
1980s–2000s
Hinge-free bending
Test jets reshape flexible wings smoothly, without gaps.

Sources: Smithsonian: Wright wing warping · NASA X-5 fact sheet · NASA NTRS 20160000937 · NASA X-53 · AFRL ACTE (2015) · Airbus AlbatrossONE (2020) · NASA MADCAT (2019) *Photos: Wright brothers (PD), Bernard Spragg (CC0), NASA (PD)*

Future scope

Where morphing-wing research is heading, and why it matters for sustainable aviation

- Programmable lattice wings
- Wings built from small lattice blocks change shape in flight; wind-tunnel tested in 2017 (NASA/MIT).
- Longer wings with free wingtips
- Albatross-style tips could take aspect ratio from 9–10 to about 18 and span past 50 m (Airbus).
- Sensing and closed-loop control
- Gust sensors let the wing react before the load arrives; modelling and control remain the main gap.



Path to net-zero CO_2 by 2050 (IATA)

65%	19%	13%	3%
sustainable fuel	offsets, capture	new technology	operations

Induced drag is about 40% of a large airliner's drag, so wing shape is a direct lever on fuel (Airbus).

Sources: NASA: What is MADCAT? (2019) · Airbus AlbatrossONE (2020) · Airbus XPW (2023) · Li et al., Prog. Aerosp. Sci. 100 (2018) · IATA Fly Net Zero

Conclusion: Key Takeaways

We started with a simple problem. A conventional wing is fixed in shape, so it is only ever ideal for one moment of a flight, and its hinges, gaps and flaps add drag, noise and weight. Birds solved this long ago. An owl's feathers, a swift's swept wings and an albatross's long wings all show that a wing can change to suit the moment. Engineers have slowly borrowed those ideas, from the Wright brothers twisting their wingtips like buzzards to seamless flaps and free-flapping wingtips tested in flight today. The gains are real, but so is the price in weight, power and complexity, and that is why airliners still fly mostly on fixed wings. The question for the future may not be whether we can copy a bird, but whether we can build a wing that senses, adapts and carries load as one system, the way a bird does without thinking. If we can, the next big step in cleaner flight may come from changing the shape of the wing itself, not only the fuel it burns.

Nature shows the target. The engineering is in carrying the load.

Sources: Smithsonian (Wright) · Bolonkin & Gilyard, NASA/TM-1999-206586 · Miller et al., NASA NTRS 20160000937 · Airbus AlbatrossONE (2020) · IATA Fly Net Zero · A350/787 camber: see slide 32

Thank You